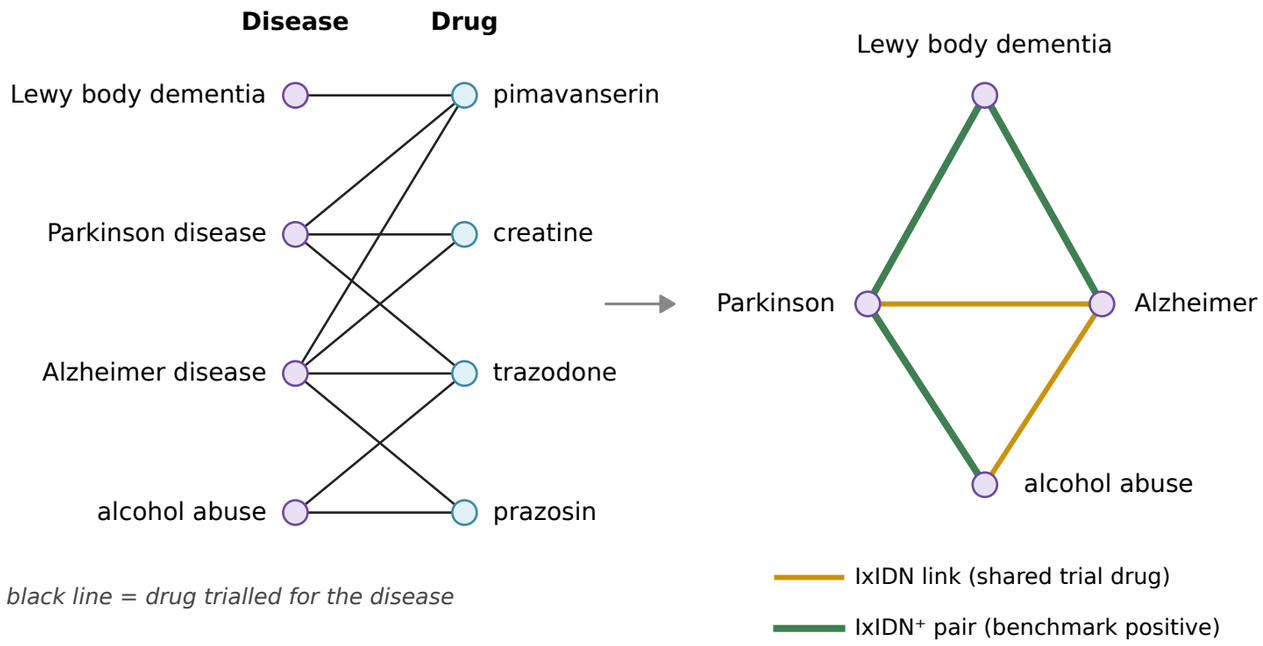
a IxIDN: trial graph projected onto diseases



b From IxIDN to IxIDN-Matched

IxIDN

1,379 diseases · 43,112 links

Evaluation universe

985 diseases

27,415 IxIDN⁺ | 155,375 IxIDN⁻ candidates

(IxIDN⁻: ≥3 steps apart, no shared trial drug with a known mechanism)

before matching, degree sum alone: AUROC 0.947

Match on 7 keys

1 degree decile, lower-degree disease
2 degree decile, higher-degree disease
3 shared organ system
4 shared Orphanet clinical area
5 name similarity (quartile)
6 MONDO similarity (quartile)
7 gene-coverage category

Language-model QC

weak shared-drug evidence dropped

IxIDN⁺

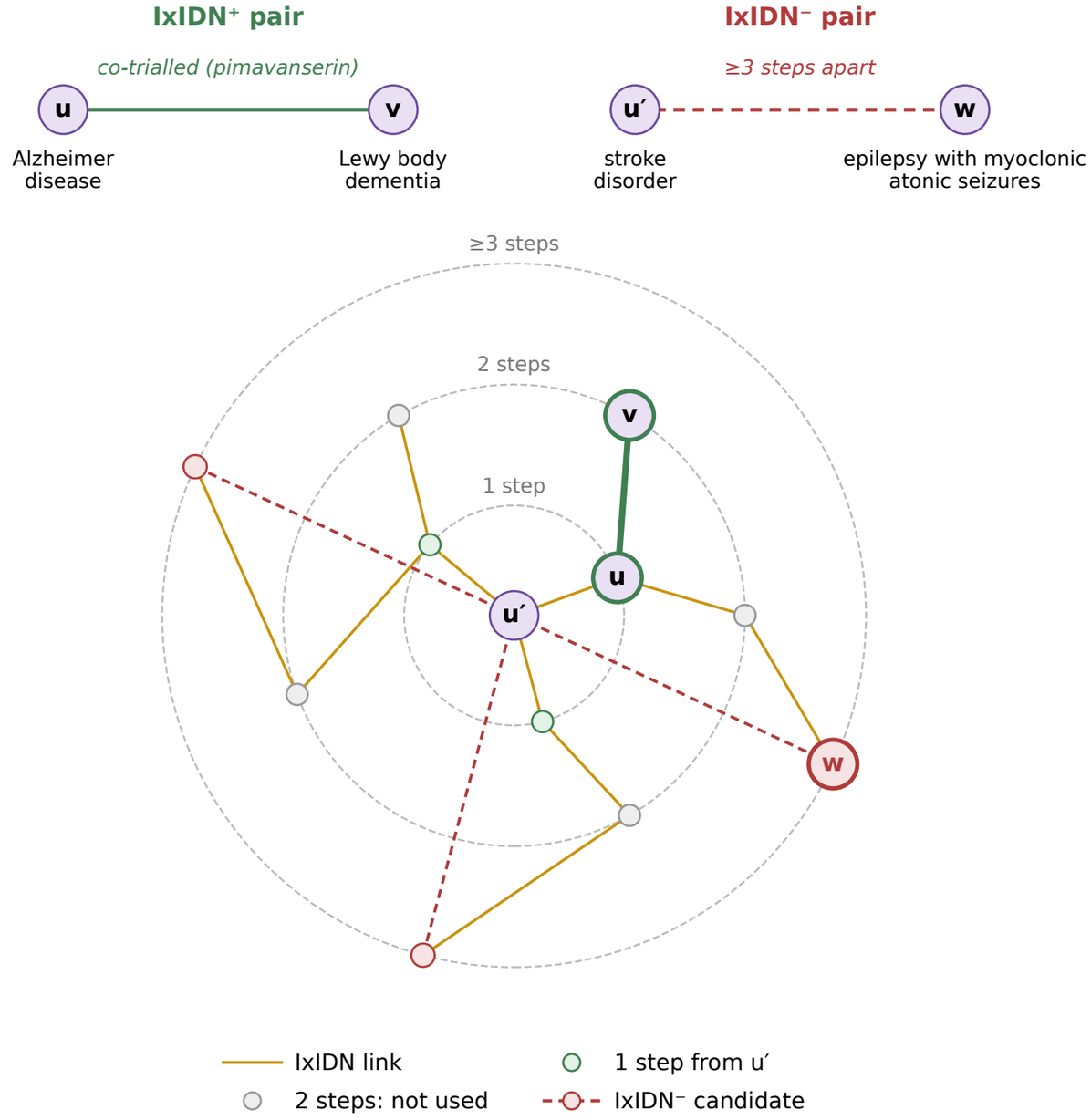
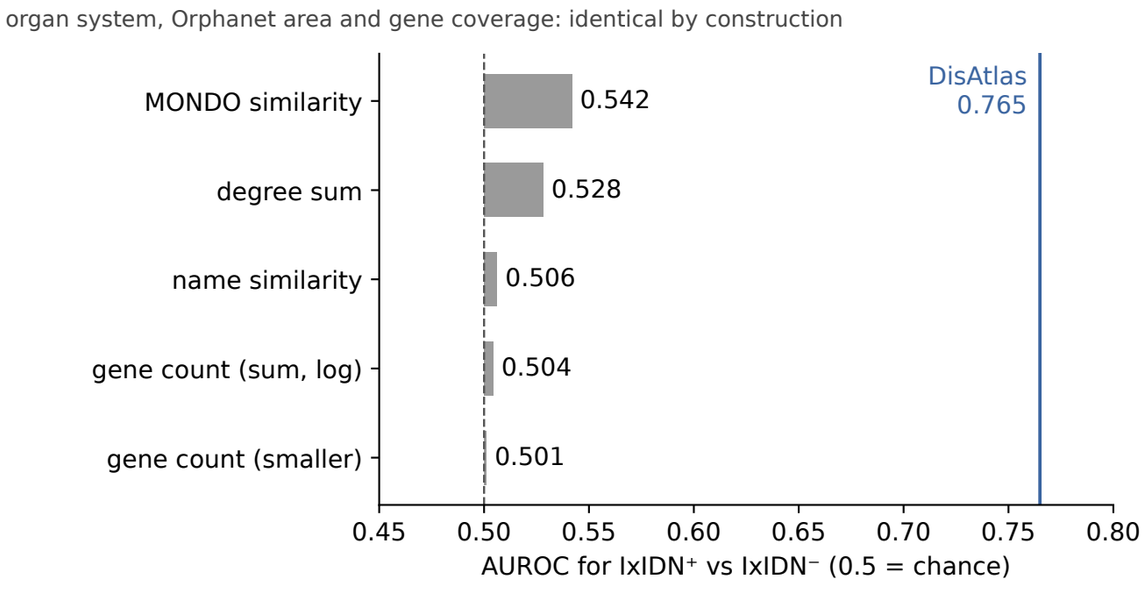
2,182 pairs

IxIDN⁻

2,182 pairs

IxIDN-Matched: 4,364 pairs over 938 diseases

c How an IxIDN⁻ pair is chosen



d Matched properties alone do not separate positives from negatives